P2 Chapter 11: Integration

Using Standard Functions

SKILL #1: Integrating Standard Functions

There's certain results you should be able to integrate straight off, by just thinking about the opposite of differentiation.

y	$\int y\,dx$
x^n	$\frac{1}{n+1}x^{n+1} + C$
e^x	$e^x + C$
$\frac{1}{x}$	$\ln\lvert x\rvert + C$
$\cos x$	$\sin x + C$
$\sin x$	$-\cos x + C$
$\sec^2 x$	$\tan x + C$
$\sec x \tan x$	$\sec x + C$
$\operatorname{cosec}^2 x$	$-\cot x + C$
$\operatorname{cosec} x \operatorname{cotan} x$	$-\operatorname{cosec} x + C$

The $|x|$ has to do with problems when x is negative (when $\ln x$ is not defined)

Remember the memorisation trick of picturing sin above cos from C3, so that 'going down' is differentiating and 'going up' is integrating, and we change the sign if the wrong way round.

It's vital you remember this one.

Have a good stare at this slide before turning your paper over – let's see how many you remember…

Quickfire Questions (without cheating!)

$$\int \sec x \tan x \ \ dx = \boxed{\quad ? \quad}$$

$$\int \sin x \ \ dx = \boxed{\quad ? \quad}$$

$$\int cosec^2 \ x \ dx = \boxed{\quad ? \quad}$$

$$\int -\cos x \ \ dx = \boxed{\quad ? \quad}$$

Quickfire Questions (without cheating!)

$$\int \sec x \tan x \ \ dx = \sec x + C$$

$$\int \sin x \ \ dx = -\cos x + C$$

$$\int cosec^2 \ x \ dx = -\cot x + C$$

$$\int -\cos x \ \ dx = -\sin x + C$$

Quickfire Questions (without cheating!)

$$\int \sec^2 x \; dx = \boxed{\quad ? \quad}$$

$$\int cosec\, x \cot x \; dx = \boxed{\quad ? \quad}$$

$$\int \frac{1}{x} \; dx = \boxed{\quad ? \quad}$$

$$\int -\sin x \; dx = \boxed{\quad ? \quad}$$

Quickfire Questions (without cheating!)

$$\int \sec^2 x \ \ dx = \tan x + C$$

$$\int cosec\ x \cot x \ \ dx = -cosec\ x + C$$

$$\int \frac{1}{x} \ dx = \ln |x| + C$$

$$\int -\sin x \ \ dx = \cos x + C$$

Quickfire Questions (without cheating!)

$$\int cosec^2\, x\, dx = \boxed{\quad ? \quad}$$

$$\int \sin x\ \ dx = \boxed{\quad ? \quad}$$

$$\int \sec x \tan x\ \ dx = \boxed{\quad ? \quad}$$

$$\int \cos x\ \ dx = \boxed{\quad ? \quad}$$

Quickfire Questions (without cheating!)

$$\int cosec^2\, x\, dx = -\cot x + C$$

$$\int \sin x\ \ dx = -\cos x + C$$

$$\int \sec x \tan x\ \ dx = \sec x + C$$

$$\int \cos x\ \ dx = \sin x + C$$

Test Your Understanding

$$\int 2\cos x + \frac{3}{x} - \sqrt{x}\, dx = \boxed{?}$$

$$\int \frac{\cos x}{\sin^2 x}\, dx = \boxed{?}$$

Hint: What 'reciprocal' trig functions does this simplify to?

[Textbook] Given that $\int_a^{3a} \frac{2x+1}{x}\, dx = \ln 12$, find the exact value of a.

?

Important Notes:
We can simplify:

$$\frac{x+1}{x} \equiv \frac{x}{x} + \frac{1}{x} \equiv 1 + \frac{1}{x}$$

However, it is **NOT** true that:

$$\frac{x}{x+1} \equiv \frac{x}{x} + \frac{x}{1}$$

It is common for students to fail to spot when they can split up a fraction to then integrate!

Test Your Understanding

$$\int 2\cos x + \frac{3}{x} - \sqrt{x}\, dx = 2\sin x + 3\ln|x| - \frac{2}{3}x^{\frac{3}{2}} + C$$

$$\int \frac{\cos x}{\sin^2 x}\, dx = \int \operatorname{cosec} x \cot x \;\; dx = -cosec\ x + C$$

Hint: What 'reciprocal' trig functions does this simplify to?

[Textbook] Given that $\int_a^{3a} \frac{2x+1}{x}\, dx = \ln 12$, find the exact value of a.

$$\int_a^{3a} 2 + \frac{1}{x}\, dx = [2x + \ln x]_a^{3a}$$
$$= (6a + \ln 3a) - (2a + \ln a)$$
$$= 4a + \ln\left(\frac{3a}{a}\right) = 4a + \ln 3$$
$$4a + \ln 3 = \ln 12$$
$$4a = \ln 12 - \ln 3 = \ln 4$$
$$a = \frac{1}{4}\ln 4$$

Important Notes:
We can simplify:
$$\frac{x+1}{x} \equiv \frac{x}{x} + \frac{1}{x} \equiv 1 + \frac{1}{x}$$
However, it is **NOT** true that:
$$\frac{x}{x+1} \equiv \frac{x}{x} + \frac{x}{1}$$
It is common for students to fail to spot when they can split up a fraction to then integrate!

Exercise 11.1

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Homework Exercise

1 Integrate the following with respect to x.

a $3\sec^2 x + \frac{5}{x} + \frac{2}{x^2}$

b $5e^x - 4\sin x + 2x^3$

c $2(\sin x - \cos x + x)$

d $3\sec x \tan x - \frac{2}{x}$

e $5e^x + 4\cos x - \frac{2}{x^2}$

f $\frac{1}{2x} + 2\operatorname{cosec}^2 x$

g $\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3}$

h $e^x + \sin x + \cos x$

i $2\operatorname{cosec} x \cot x - \sec^2 x$

j $e^x + \frac{1}{x} - \operatorname{cosec}^2 x$

2 Find the following integrals.

a $\int\left(\frac{1}{\cos^2 x} + \frac{1}{x^2}\right) dx$

b $\int\left(\frac{\sin x}{\cos^2 x} + 2e^x\right) dx$

c $\int\left(\frac{1+\cos x}{\sin^2 x} + \frac{1+x}{x^2}\right) dx$

d $\int\left(\frac{1}{\sin^2 x} + \frac{1}{x}\right) dx$

e $\int \sin x(1 + \sec^2 x)\, dx$

f $\int \cos x(1 + \operatorname{cosec}^2 x)\, dx$

g $\int \operatorname{cosec}^2 x(1 + \tan^2 x)\, dx$

h $\int \sec^2 x(1 - \cot^2 x)\, dx$

i $\int \sec^2 x(1 + e^x \cos^2 x)\, dx$

j $\int\left(\frac{1+\sin x}{\cos^2 x} + \cos^2 x \sec x\right) dx$

3 Evaluate the following. Give your answers as exact values.

a $\int_3^7 2e^x\, dx$

b $\int_1^6 \frac{1+x}{x^3}\, dx$

c $\int_{\frac{\pi}{2}}^{\pi} -5\sin x\, dx$

d $\int_{-\frac{\pi}{4}}^{0} \sec x(\sec x + \tan x)\, dx$

Watch out When applying limits to integrated trigonometric functions, always work in radians.

Homework Exercise

4 Given that a is a positive constant and $\int_a^{2a} \frac{3x-1}{x}\,dx = 6 + \ln\left(\frac{1}{2}\right)$, find the exact value of a. **(4 marks)**

5 Given that a is a positive constant and $\int_{\ln 1}^{\ln a} e^x + e^{-x}\,dx = \frac{48}{7}$, find the exact value of a. **(4 marks)**

6 Given $\int_2^b (3e^x + 6e^{-2x})\,dx = 0$, find the value of b. **(4 marks)**

7 $f(x) = \frac{1}{8}x^{\frac{3}{2}} - \frac{4}{x}$, $x > 0$

a Solve the equation $f(x) = 0$. **(2 marks)**

b Find $\int f(x)\,dx$. **(2 marks)**

c Evaluate $\int_1^4 f(x)\,dx$, giving your answer in the form $p + q\ln r$, where p, q and r are rational numbers. **(3 marks)**

Homework Answers

1 **a** $3\tan x + 5\ln|x| - \frac{2}{x} + c$ **b** $5e^x + 4\cos x + \frac{x^4}{2} + c$

c $-2\cos x - 2\sin x + x^2 + c$ **d** $3\sec x - 2\ln|x| + c$

e $5e^x + 4\sin x + \frac{2}{x} + c$ **f** $\frac{1}{2}\ln|x| - 2\cot x + c$

g $\ln|x| - \frac{1}{x} - \frac{1}{2x^2} + c$ **h** $e^x - \cos x + \sin x + c$

i $-2\operatorname{cosec} x - \tan x + c$ **j** $e^x + \ln|x| + \cot x + c$

2 **a** $\tan x - \frac{1}{x} + c$ **b** $\sec x + 2e^x + c$

c $-\cot x - \operatorname{cosec} x - \frac{1}{x} + \ln|x| + c$

d $-\cot x + \ln|x| + c$ **e** $-\cos x + \sec x + c$

f $\sin x - \operatorname{cosec} x + c$ **g** $-\cot x + \tan x + c$

h $\tan x + \cot x + c$ **i** $\tan x + e^x + c$

j $\tan x + \sec x + \sin x + c$

3 **a** $2e^7 - 2e^3$ **b** $\frac{95}{72}$ **c** -5 **d** $2 - \sqrt{2}$

4 $a = 2$

5 $a = 7$

6 $b = 2$

7 **a** $x = 4$ **b** $\frac{1}{20}x^{\frac{5}{2}} - 4\ln|x| + c$

c $\frac{31}{20} - 4\ln 4$